APPLICATION On Behalf of

Shanghai Qiaohe Industrial Co.，Ltd

Date of Test: **Apr.10.2026~Apr.17.2026**
Date of Report: **Apr.10.2026**
Report Number: **STUESO0520024332206LM**

TEST REPORT

US IEC 60335-1:2010

Household and similar electrical appliances -Safety -

Part 1: General requirements

US IEC 60335-2-7:2012

Household and similar electrical appliances - Safety - Part 2-7: Particular requirements for washing machines

Tested by (+ signature). .....:	Alice
Approved by (+ signature).....:	Alisa
Date of issue .....:	Apr.17.2026
Content..... .....:	83Pages

Testing laboratory.....:	Guangzhou STU Hengchuang Testing Service Co. Ltd
Address .....:	No.202, Building A, Jingye Sanjie, Yushu Industrial Park, Guangzhou Economic & Technology Development Zone, Guangzhou. Guangdong., China
Testing location .....:	Same as above

Applicant .....:	Shanghai Qiaohe Industrial Co.，Ltd
Address.....:	No. 699, Caosheng Road, Jiading District, Shanghai, China

Standard.....:	US IEC 60335-2-7: 2012 (Also see US IEC 60335-1: 2010)
Test procedure…….....…:	IEC test report
Non-standard test method….....:	N/A

Test Report Forms No. .....:	US IEC 60335-2-7: 2012 (Also see US IEC 60335-1: 2010)

Test item .....:	washing machine
Trade Mark.....	SIMPLECLEAN
Model and/or Type .....:	EC/27FW
Rating(s)…….....……:	240V 50Hz 400W

Test item particulars
Classification of installation and use......: Class I
Supply Connection …….....: Type Y
Possible test case verdicts:
Test case does not apply to the test object .....: N(N/A)
Test item does meet the requirement .....: P(Pass)
Test item does not meet the requirement. .....: F(Fail)
Testing
Date of receipt of test item……….....….: Apr.10.2026
Date(s) of performance of test………...…..: Apr.10.2026~Apr.17.2026
General remarks:
This report shall not be reproduced except in full without the written approval of the testing laboratory. The test results presented in this report relate only to the item(s) tested. " (see remark #)" refers to a remark appended to the report. "(see Annex #)" refers to an annex appended to the report. Clause numbers between brackets refer to clauses in US IEC 60335-2-7:2012 (Also see US IEC 60335-1: 2010) Throughout

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
5	GENERAL CONDITIONS FOR THE TESTS		/
	Tests performed according to cl. 5, e.g. nature of supply, sequence of testing, etc.		P
6	CLASSIFICATION		/
6.1	Protection against electric shock: Class 0, 0I, I, II, III..... ..... :	Class I	P
6.2	Protection against harmful ingress of water		P
7	MARKING AND INSTRUCTIONS		/
7.1	Rated voltage or rated voltage range in volts... :	240V	P
	Symbol for nature of supply, unless the rated frequency is marked..... :	50Hz	P
	Manufacturer's or responsible vendor's name, trademark or identification mark	See page one	P
	Model or type reference	See labels	P
	Rated power input in watts or rated current in amperes..... :	400W	P
	Symbol IEC 60417-5172 (2003-02) for class II appliances only		N/A
	IP number according to degree of protection against ingress of water, other than IPXO. :		N/A
	Symbol IEC 60417-5180 (2003-02). for class III appliances.		N/A
	This marking is not necessary for appliances that are operated only by batteries (primary batteries or secondary batteries that are recharged outside of the appliance).		N/A
	Appliances intended partially immersed in water for cleaning marked with maximum level of immersion and (IEC 60335-2-7)		P
	with substance of following: Do not immerse beyond this level. (IEC 60335-2-7)		P

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
	When the provisions of footnote b to Table 101 apply, the appliance shall be marked with: (IEC 60335-2-7)		P
	-the substance of “CAUTION: Hot surface”, or (IEC 60335-2-7)		P
	-symbol IEC 60417-5041 (2002-10). (IEC 60335-2-7)		P
7.2	Stationary appliances for multiple supply shall be marked with the substance of the following: Warning: Before obtaining access to terminals, all supply circuits must be disconnected. This warning shall be placed in the vicinity of the terminal cover.		P
7.3	Appliances having a range of rated values and which can be operated without adjustment throughout the range shall be marked with the lower and upper limits of the range separated by a hyphen.		N/A
7.5	For appliances marked with more than one rated voltage or with one or more rated voltage ranges. the rated power input or rated current for each of these voltages or ranges shall be marked. However, if the difference between the limits of a rated voltage range does.1 not exceed 10 % of the arithmetic mean value of the range, the marking for rated power input or rated current may be related to the arithmetic mean value of the range.		N/A
	The upper and lower limits of the rated power input or rated current shall be marked on the appliance so that the relation between input and voltage is clear.		N/A
7.6	[symbol IEC 60417-5041 (2002-10)] caution, hot surface (IEC 60335-2-7)		P

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
7.7	Appliances to be connected to more than two supply conductors and appliances for multiple supply shall have a connection diagram fixed to them, unless the correct mode of connection is obvious.		N/A
7.9	Unless it is obviously unnecessary, switches which may give rise to a hazard when .operated shall be marked or placed so as to indicate clearly which part of the appliance they control. Indications used for this purpose shall, wherever practicable, be comprehensible without a knowledge of languages or national standards..		N/A
7.10	The different positions of switches on stationary appliances and the different positions of controls on all appliances shall be indicated by figures, letters or other visual means. This requirement also applies to switches which are part of a control.	1,2,3,4	P
	The figure 0 shall not be used for any other indication unless it is positioned and associated with other numbers so that it does not give rise to confusion with the indication of the off position.	OFF	N/A
7.11	Controls intended to be adjusted during installation or in normal use shall be provided., with an indication for the direction of adjustment. NOTE An indication of + and- is considered to be sufficient		N/A
7.12	Instructions for appliances incorporating appliance inlet, and intended partially or completely immersed in water for cleaning, state that connector removed before appliance cleaned and appliance inlet dried before appliance used again (IEC 60335-2-7)		P
	Instructions include substance of following (IEC 60335-2-7):		P

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
	This appliance is intended to be used in household and similar applications such as: (IEC 60335-2-7)		P
	-staff kitchen areas in shops, offices and other working environments; (IEC 60335-2-7)		P
	-farm houses; (IEC 60335-2-7)		P
	- by clients in hotels, motels and other residential type environments; (IEC 60335-2-7)		P
	-bed and breakfast type environments. (IEC 60335-2-7)		P
	Instructions for appliances intended used with connector incorporating thermostat state that only appropriate connector used (IEC 60335-2-7)		P
	Instructions for appliances with surfaces of glass- ceramic or similar material forms part of enclosure of live parts include substance of following: WARNING: Do not use the appliance if the surface is cracked.(IEC 60335-2-7)		P
	Instructions for appliances used with particular vessels not supplied specify vessels used (IEC 60335-2-7)		P
	If symbol IEC 60417-5041 (2002-10) is marked on appliances, its meaning shall be explained and location of areas where the limits are exceeded shall be stated or shown by a drawing. (IEC 60335-2-7)		P

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
7.12.4	The instructions for built-in appliances shall include information with regard to the following: -- dimensions of the space to be provided for the appliance; -- dimensions and position of the means for supporting and fixing the appliance within this space; -- minimum distances between the various parts of the appliance and the surrounding structure; -- minimum dimensions of ventilating openings and their correct arrangement; -- connection of the appliance to the supply mains and the interconnection of any separate components; -- necessity to allow disconnection of the appliance from the supply after installation, unless the appliance incorporates a switch complying with 24.3. The disconnection may be achieved by having the plug accessible or by incorporating a switch in the fixed wiring in accordance with the wiring rules.		N/A
7.12.7	The instructions for fixed appliances shall state how the appliance is to be fixed to.. its support. The method of fixing stated is not to depend on the use of adhesives since they are not considered to be a reliable fixing means.		N/A
7.12.8	The instructions for appliances connected to the water mains shall state -- the maximum inlet water pressure, in pascals; -- the minimum inlet water pressure, in pascals, if this is necessary for the correct operation of the appliance.		N/A

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
	The instructions for appliances connected to the water mains by detachable hose-sets shall state that the new hose-sets supplied with the appliance are to be used and that old hose-sets should not be reused.		N/A
7.13	Instructions and other text required by this standard shall be written in an official language of the country in which the appliance is to be sold.		N/A
7.14	The markings requiredHeight of the triangle in symbol IEC 60417-5041 (2002-10) is at least 10 mm. (IEC 60335-2-7)		P
7.15	The markings specified in 7.1 to 7.5 shall be on a main part of the appliance.		N/A
	Operation of the appliance as in normal use: marking specified for hot surfaces is clearly visible and (IEC 60335-2-7)		P
	it is not placed on a hot functional surface or on surfaces that are exempted from the limits of Table 101. (IEC 60335-2-7)		P
8	PROTECTION AGAINST ACCESS TO LIVE PARTS		/
8.1	Appliances shall be constructed and enclosed so that there is adequate protection against accidental contact with live parts.		P
	Compliance is checked by inspection and by the tests of 8.1.1 to 8.1.3, as applicable, taking into account 8.1.4 and 8.1.5.		P
8.1.1	The requirement of 8.1 applies for all positions of the appliance when it is operated as in normal use, and after the removal of detachable parts		P

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
	lamps located behind a detachable cover are not removed, provided that the appliance can be isolated from the supply mains by means of a plug or an all-pole switch. However, during insertion or removal of lamps which are located behind a detachable cover, protection against contact with live parts of the lamp cap shall be ensured.		N/A
	Test probe B of IEC 61032 is applied with a force not exceeding 1 N. the appliance being in every possible position except that appliances normally used on the floor and having a mass exceeding 40 kg are not tilted.	Meet the requirements	P
	Through openings, the test probe is applied to any depth that the probe will permit and is rotated or angled before, during and after insertion to any position.		P
8.1.2	Test probe 13 of IEC 61032 is applied with a force not exceeding 1 n through openings . in class 0 appliances, class II appliances and class II constructions, except for those giving access to lamp caps and live parts in socket-outlets.		N/A
8.1.3	Instead of test probe B and test probe 13, for appliances other than those of class II test probe 41 of IEC 61032 is applied with a force not exceeding 1 n to live parts of visibly glowing heating elements. all poles of which can be disconnected by a single switching.. action. It is also applied to parts supporting such elements, provided that it is obvious from the outside of the appliance, without removing covers and similar parts, that these supporting parts are in contact with the element.		P
8.1.5	Live parts of built-in appliances, fixed appliances and appliances delivered in separate units, shall be protected at least by basic insulation before installation or assembly.		N/A

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
8.2	Class II appliances and class II constructions shall be constructed and enclosed so that there is adequate protection against accidental contact with basic insulation and metal parts separated from live parts by basic insulation only.		N/A
	It shall only be possible to touch parts which are separated from live parts by double insulation or reinforced insulation.		N/A
9	STARTING OF MOTOR-OPERATED APPLIANCES		/
	NOTE: Requirements and tests are specified in part 2 when necessary.		N/A
10	POWER INPUT AND CURRENT		/
10.1	If an appliance is marked with rated power input, the power input at normal operating temperature shall not deviate from the rated power input by more than the deviation shown in Table 1.	(see appended table)	N/A
	The deviation for motor-operated appliances applies for combined appliances if the power.. input of the motors is more than 50 % of the rated power input. The permissible deviations apply for both limits of the range for appliances marked with a rated voltage range having limits differing by more than 10 % of the arithmetic mean value of the range.		N/A
	The test is carried out at both the upper and lower limits of the ranges for appliances marked with one or more rated voltage ranges.		N/A
10.2	If an appliance is marked with rated current, the current at normal operating temperature shall not deviate from the rated current by more than the deviation shown in Table 2.	(see appended table)	N/A

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
	The deviation for motor-operated appliances applies for combined appliances if the current of the motors is more than 50 % of the rated current. The permissible deviations apply for both limits of the range for appliances marked with a rated voltage range having limits differing by more than 10 % of the arithmetic mean value of the range.		N/A
11	HEATING		N/A
11.1	Appliances and their surroundings shall not attain excessive temperatures in normal use.		N/A
11.3	Temperature rises, other than those of windings. are determined by means of fine-wire thermocouples positioned so that they have minimum effect on the temperature of the part under test.		N/A
	Thermocouples used for determining the temperature rise of the surface of walls, ceiling and floor of the test corner are attached to the back of small blackened disks of copper or brass, 15 mm in diameter and 1 mm thick. The front of the disk is flush with the surface of the board.		N/A
	The temperature rise of electrical insulation, other than that of windings. is determined on the surface of the insulation at places where failure could cause -- a short circuit; -- contact between live parts and accessible metal parts; -- bridging of insulation; -- a reduction of clearances or creepage distances below the values specified in Clause 29.		N/A

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
	Temperature rises of windings are determined by the resistance method unless the windings are non-uniform or if it is difficult to make the necessary connections, in which case the temperature rise is determined by means of thermocouples.		N/A
11.4	Heating appliances are operated under normal operation and at 1.15 times rated power input..		N/A
11.5	Motor-operated appliances are operated under normal operation and supplied with the most unfavourable voltage between 0,94 times and 1,06 times the rated voltage.....:	1.06×240V=254.4V	P
11.6	Combined appliances are operated under normal operation and supplied with the most unfavourable voltage between 0,94 times and 1,06 times the rated voltage.....:		N/A
11.8	During the test, the temperature rises are monitored continuously and shall not exceed the values shown in Table 3.		N/A
	The test results	The temperature is not out of tolerance	P
13	LEAKAGE CURRENT AND ELECTRIC STRENGTH AT OPERATING TEMPERATURE		/
13.1	At operating temperature, the leakage current of the appliance shall not be excessive and its electric strength shall be adequate.		P
	The appliance is operated under normal operation for the duration specified in 11.7.		P
	Heating appliances are operated at 1,15 times the rated power input.		N/A
	Motor-operated appliances and combined appliances are supplied at 1.06 times rated voltage.		P

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
	Three-phase appliances, which according to the instructions for installation are also suitable . for single-phase supply, are tested as single-phase appliances with the three circuits connected in parallel.		P
	Protective impedance and radio interference filters are disconnected before carrying out the tests.		P
13.2	The leakage current is measured by means of the circuit described in Figure 4 of IFC 60990:1999. For class 0l appliances and class I appliances, except parts . of class II construction, C may be replaced by a low impedance ammeter responding to the, rated frequency of the appliance.	(see appended table)	P
	The leakage current is measured between any pole of the supply and accessible metal parts connected to metal foil having an area not exceeding 20 cm x 10 cm which is in contact with accessible surfaces of insulating materials.		P
	The metal foil has the largest area possible on the surface under test without exceeding the dimensions specified. If its area is smaller than the surface under test. it is moved to test all parts of the surface. The heat dissipation of the appliance is not to be affected by the metal foil.		P
	For combined appliances. the total leakage current may be within the limits specified for heating appliances or motor-operated appliances, whichever is the greater, but the two limits are not added.		N/A
13.3	The appliance is disconnected from the supply and the insulation is immediately subjected to a voltage having a frequency of 50 Hz or 60 Hz for 1 min. in accordance with IEC 61180-1.		P

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
	The high-voltage source used for the test is to be capable of supplying a short circuit current I between the output terminals after the output voltage has been adjusted to the appropriate test voltage. The overload release of the circuit is not to be operated by any current below the tripping current I. The values of I, and I, are given in Table 5 for various high-voltage sources.	(see appended table)	P
	The test voltage is applied between live parts and accessible parts, non-metallic parts being covered with metal foil. For class II constructions having intermediate metal between live parts and accessible parts, the voltage is applied across the basic insulation and the supplementary insulation.		P
14	TRANSIENT OVERVOLTAGES		/
	Appliances shall withstand the transient over-voltages to which they may be subjected.		N/A
	Compliance is checked by subjecting each clearance having a value less than those specified in Table 16 to an impulse voltage test.	(see appended table)	N/A
	The impulse test voltage has a no-load wave shape corresponding to the 1,2/50 us standard impulse specified in IEC 61180-1. It is supplied from a generator having a conventional output impedance not exceeding 42 2. The impulse test voltage is applied three times for each polarity with intervals of at least 1s.		N/A
	The impulse test voltage is specified in Table 6 for rated impulse voltages given in Table 15.	(see appended table)	N/A
15	MOISTURE RESISTANCE		/
15.1	The enclosure of the appliance shall provide the degree of protection against moisture in accordance with the classification of the appliance.		P

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
	The appliance shall then withstand the electric strength test of 16.3 and, after carefully wiping the external enclosure to remove any surplus water. an inspection shall show that there is no trace of water on insulation which could result in a reduction of clearances or creepage distances below the values specified in Clause 29.		P
15.1.1	Appliances other than those classified IPXO are subjected to the tests of IEC 60529 as follows. -- IPX1 appliances as described in Subclause 14.2.1; -- IPX2 appliances as described in Subclause 14.2.2; -- IPX3 appliances as described in Subclause 14.2.3a;		P
	-- IPX4 appliances as described in Subclause 14.2.4a; -- IPX5 appliances as described in Subclause 14.2.5; -- IPX6 appliances as described in Subclause 14.2.6; -- IPX7 appliances as described in Subclause 14.2.7. For this test the appliance is immersed in water containing approximately 1 % NaCl		P
	Water valves containing live parts and that are incorporated in external hoses for connection of an appliance to the water mains are subjected to the test specified for IPX7 appliances.		N/A
15.1.2	Hand-held appliances are turned continuously through the most unfavourable positions during the test.		N/A
	Built-in appliances are installed in accordance with the instructions.		N/A

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
	Appliances normally fixed to a wall and appliances with pins for insertion into socket-outlets: are mounted as in normal use in the centre of a wooden board having dimensions which are 15 cm ±5 cm in excess of those of the orthogonal projection of the appliance on the board.The wooden board is placed at the centre of the oscillating tube.		N/A
	For IPX3 appliances, the base of wall-mounted appliances is placed at the same level as the pivot axis of the oscillating tube.		N/A
	For IPX4 appliances. the horizontal centre line of the appliance is aligned with the pivot axis of the oscillating tube. However, for appliances normally used on the floor or table. the movement is limited to two times 900 from the vertical for a period of 5 min, the support being placed at the level of the pivot axis of the oscillating tube.		N/A
	Appliances normally fixed to a ceiling are mounted underneath a horizontal unperforated support that is constructed to prevent water spraying onto its top surface. The pivot axis of the oscillating tube is located at the same level as the underside of the support and aligned centrally with the appliance. The spray is directed upwards. For IPX4 appliances, the movement of the tube is limited to two times 909 from the vertical for a period of 5 min.		N/A
	Appliances with type X attachment, except those having a specially prepared cord, are fitted with the lightest permissible type of flexible cord of the smallest cross-sectional area specified in Table 13.		N/A

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
	Detachable parts are removed and subjected, if necessary. to the relevant treatment with the main part. However. if the instructions state that a part has to be removed for user maintenance and a tool is needed. this part is not removed.		N/A
15.2	Appliances shall be constructed so that spillage of liquid in normal use does not affect their electrical insulation.		P
15.3	Appliances shall be proof against humid conditions that may occur in normal use.		P
	Appliances that were subjected to the tests of 15.1 or 15.2 are placed in normal ambient conditions for 24 h.		P
	Cable entries, if any, are left open. If knock-outs are provided, one of them is opened.Detachable parts are removed and subjected, if necessary, to the humidity test with the main part.		P
	The humidity test is carried out for 48 h in a humidity cabinet containing air with a relative humidity of (93 ± 3) %. The temperature of the air is maintained within 2 K of any convenient value t between 20 C and 30 C. Before being placed in the humidity cabinet, the appliance is brought to a temperature of t+d c.	94%	P
	The appliance shall then withstand the tests of Clause 16 in the humidity cabinet or in the room in which the appliance was brought to the prescribed temperature after reassembly of those parts that may have been removed.		P
16	LEAKAGE CURRENT AND ELECTRIC STRENGTH		/
16.1	The leakage current of the appliance shall not be excessive and its electric strength shall be adequate.		P
	Protective impedance is disconnected from live parts before carrying out the tests.		N/A

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
	The tests are carried out on the appliance at room temperature and not connected to the supply mains.		P
16.2	An a.c. test voltage is applied between live parts and		P
	-- accessible metal parts that are intended to be connected to protective earth, for class T appliances and class 0I appliances; -- metal foil having an area not exceeding 20 cm x 10 cm which is in contact with accessible surfaces of insulating materials and metal parts not intended to be connected to protective earth, for class 0 appliances, class II appliances, class II constructions and class III appliances.		P
	The test voltage is -- 1,06 times rated voltage, for single-phase appliances; -- 1,06 times rated voltage, divided by 3, for three-phase appliances. The leakage current is measured within 5 s after the application of the test voltage.	1.06×240V=254.4V Leakage current≤3.5mA	P
	The values specified above are doubled if all controls have an off position in all poles. They are also doubled if		P
	-- the appliance has no control other than a thermal cut-out, or -- all thermostats, temperature limiters and energy regulators do not have an off position, or -- the appliance has radio interference filters. In this case, the leakage current with the filter disconnected shall not exceed the limits specified.		P

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
	For combined appliances, the total leakage current may be within the limits specified for heating appliances or motor-operated appliances, whichever is the greater, but the two limits are not added.		N/A
16.3	Immediately after the test of 16.2. the insulation is subjected to a voltage having a . frequency of 50 Hz or 60 Hz for 1 min in accordance with IEC 61180-1. The values of the test voltage for different types of insulation are given in Table 7. Accessible parts of insulating material are covered with metal foil.		P
	A test voltage is applied between accessible metal parts and the supply cord which is wrapped with metal foil at the point where the supply cord is located in an inlet bushing and where, for appliances having a type X attachment, the supply cord is located in a cord guard or a cord anchorage with their clamping screws, if any, being tightened to two-thirds of the torque specified in Table 14. The test voltage is 1 250 V for class 0 appliances and class Iappliances and 1 750 V for class II appliances.		P
17	OVERLOAD PROTECTION OF TRANSFORMERS AND ASSOCIATED CIRCUITS		/
	Appliances incorporating circuits supplied from a transformer shall be constructed so that in the event of short circuits which are likely to occur in normal use, excessive temperatures do not occur in the transformer or in the circuits associated with the transformer.		N/A

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
	Compliance is checked by applying the most unfavourable short circuit or overload which is likely to occur in normal use, the appliance being supplied with 1.06 times or 0.94 times rated voltage, whichever is the more unfavourable. Basic insulation is not short circuited.		N/A
	The temperature rise of the insulation of the conductors of safety extra-low voltage circuits shall not exceed the relevant value specified in Table 3 by more than 15 K		N/A
	The temperature of windings shall not exceed the values specified in Table 8. However, these limits do not apply to fail-safe transformers complying with Subclause 15.5 of IEC 61558-1.		N/A
19	ABNORMAL OPERATION		/
	Electronic circuits shall be designed and applied so that a fault condition will not render the appliance unsafe with regard to electric shock, fire hazard, mechanical hazard or dangerous malfunction.		P
	Appliances incorporating heating elements are subjected to the tests of 19.2 and 19.3. In addition, such appliances having a control that limits the temperature during the test of Clause 11 are subjected to the tests of 19.4 and, when applicable, to the test of 19.5.Appliances incorporating PTC heating elements are also subjected to the test of 19.6.		N/A
	Appliances incorporating motors are subjected to the tests of 19.7 to 19.10, as applicable.		N/A
	Appliances incorporating electronic circuits are also subjected to the tests of 19.11 and19.12, as applicable.		P
	Appliances incorporating contactors or relays are subjected to the test of 19.14.		N/A

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
	Appliances incorporating voltage selector switches are subjected to the test of 19.15.		N/A
	Unless otherwise specified, the tests are continued until a non-self-resetting thermal cut out operates or until steady conditions are established. If a heating element or an intentionally weak part becomes permanently open-circuited, the relevant test is repeated on a second sample. This second test shall be terminated in the same mode unless the test is otherwise satisfactorily completed.		N/A
	For combined appliances. the tests are carried out with motors and heating elements: operating simultaneously under normal operation, the appropriate tests being applied one at• a time to each motor and heating element.		N/A
19.2	Appliances with heating elements are tested under the conditions specified in Clause 11 but with restricted heat dissipation. The supply voltage, determined prior to the test. is that required to provide a power input of 0.85 times rated power input under normal operation when the power input has stabilized. This voltage is maintained throughout the test.		N/A
19.4	The appliance is tested under the conditions specified in Clause 11. Any control that limits the temperature during the test of Clause 11 is short-circuited.		P
19.5	The test of 19.4 is repeated on class 0l appliances and class I appliances incorporating tubular sheathed or embedded heating elements.		N/A
	This test is repeated with the polarity of the supply to the appliance reversed and with the other end of the element connected to the sheath.		N/A

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
	The test is not carried out on appliances intended to be permanently connected to fixed wiring and on appliances where an all-pole disconnection occurs during the test of 19.4.		N/A
19.6	Appliances with PTC heating elements are supplied at rated voltage until steady conditions with regard to power input and temperature are established.		N/A
	The working voltage of the PTC heating element is increased by 5 % and the appliance is operated until steady conditions are re-established. The voltage is then increased in similar steps until 1,5 times working voltage is reached, or until the PTC heating element ruptures, whichever occurs first.		N/A
19.7	The appliance is operated under stalled conditions by -- locking the rotor of appliances for which the locked rotor torque is smaller than the full load torque; -- locking moving parts of other appliances.		N/A
	Appliances incorporating motors and having capacitors in the circuit of an auxiliary winding, are operated with the rotor locked, the capacitors being open-circuited one at a time. The test is repeated with the capacitors short-circuited one at a time, unless they are of class S2 or S3 of IEC 60252-1.		N/A
	For each of the tests, appliances provided with a timer or programmer are supplied at rated voltage for a period allowed to the maximum period allowed by the timer or programmer.		N/A

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
19.8	One phase of appliances incorporating multi-phase motors is disconnected. The appliance is then operated under normal operation and supplied at rated voltage for the period specified in 19.7.		N/A
19.9	A running overload test is carried out on appliances incorporating motors that are intended to be remotely or automatically controlled or liable to be operated continuously.		N/A
	Motor-operated appliances and combined appliances for which Subclause 30.2.3 is . applicable and that use overload protective devices relying on electronic circuits to protect the motor windings, other than those that sense winding temperatures directly, are also subjected to the running overload test.		N/A
	The appliance is operated under normal operation and supplied at rated voltage until steady conditions are established.		N/A
	The load is then increased so that the current through the motor windings is raised by 10 % and the appliance is operated again until steady conditions are established, the supply voltage being maintained at its original value. The load is again increased and the test is repeated until the protective device operates or the motor stalls.		N/A
19.10	Appliances incorporating series motors are operated with the lowest possible load and supplied at 1,3 times rated voltage for 1 min.....:	1.06×240V=254.4V	P
19.11	Electronic circuits are checked by evaluation of the fault conditions specified in19.11.2 for all circuits or parts of circuits, unless they comply with the conditions specified in19.11.1.		N/A

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
	Appliances incorporating an electronic circuit that relies upon a programmable component to function correctly are subjected to the test of 19.11.4.8, unless restarting at any point in the operating cycle after interruption of operation due to a supply voltage dip will not result in a , hazard. The test is carried out after removal of all batteries and other components intended to maintain the programmable component supply voltage during mains supply voltage dips, interruptions and variations.		N/A
	Appliances having a device with an off position obtained by electronic disconnection. or a device that can place the appliance in a stand-by mode, are subjected to the tests of 19.11.4.		N/A
19.11.1	Fault conditions a) to g) specified in 19.11.2 are not applied to circuits or parts of circuits when both of the following conditions are met: -- the electronic circuit is a low-power circuit as described below;		N/A
	-- protection against electric shock. fire hazard. mechanical hazard or dangerous malfunction of other parts of the appliance does not rely on the correct functioning of the electronic circuit.		N/A
	The appliance is supplied at rated voltage and a variable resistor, adjusted to its maximum, resistance, is connected between the point to be investigated and the opposite pole of the supply source. The resistance is then decreased until the power consumed by the resistor reaches a maximum.		N/A

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
	Points closest to the supply source at which the maximum power delivered to this resistor does not exceed 15 W at the end of 5 s are called low-power points.The part of the circuit farther from the supply source than a low-power point is considered to be a low-power circuit.		N/A
19.11.3	If the appliance incorporates a protective electronic circuit which operates to ensure compliance with Clause 19, the relevant test is repeated with a single fault simulated. i as indicated in a) to g) of 19.11.2.		N/A
19.11.4	Appliances having a device with an off position obtained by electronic disconnection. or a device that can be placed in the stand-by mode, are subjected to the tests of 19.11.4.1 to 19.11.4.7. The tests are carried out with the appliance supplied at rated voltage, the device being set in the off position or in the stand-by mode.		N/A
	Appliances incorporating a protective electronic circuit are subjected to the tests of19.11.4.1 to 19.11.4.7. The tests are carried out after the protective electronic circuit has operated during the relevant tests of Clause 19 except 19.2. 19.6 and 19.11.3. However, appliances that are operated for 30 s or 5 min during the test of 19.7 are not subjected to the tests for electromagnetic phenomena.		N/A
19.11.4.1	The appliance is subjected to electrostatic discharges in accordance with . IEC 61000-4-2. test level 4 being applicable. Ten discharges having a positive polarity and ten discharges having a negative polarity are applied at each preselected point.		N/A
19.11.4.2	The appliance is subjected to radiated fields in accordance with IEC 61000-4-3.		N/A

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
19.11.4.3	The appliance is subjected to fast transient bursts in accordance with IEC 61000-4-4. Test level 3 with a repetition rate of 5 kHz is applicable for signal and control lines. Test level 4 with a repetition rate of 5 kHz is applicable for the power supply lines. The bursts are applied for 2 min with a positive polarity and for 2 min with a negative polarity.		N/A
19.11.4.4	The power supply terminals of the appliance are subjected to voltage surges in accordance with IEC 61000-4-5. five positive impulses and five negative impulses being applied at the selected points. Test level 3 is applicable for the line-to-line coupling mode. a generator having a source impedance of 2 Q being used. Test level 4 is anplicable for the line-to-earth coupling mode, a generator having a source impedance of 12 2 being used.		N/A
19.11.4.5	The appliance is subjected to injected currents in accordance with IEC 61000-4-6 test level 3 being applicable. During the test, all frequencies between 0,15 MHz to 80 MHz are covered.		N/A
19.11.4.6	For appliances having a rated current not exceeding 16 A, the appliance is , subjected to the class 3 voltage dips and interruptions in accordance with IEC 61000-4-11.The values specified in Table 1 and Table 2 of IFC 61000-4-11 are applied at zero crossing of the supply voltage.		N/A
	For appliances having a rated current exceeding 16 A. the appliance is subjected to the class 3 voltage dips and interruptions in accordance with IEC 61000-4-34. The values specified in Table 1 and Table 2 of IEC 61000-4-34 are applied at zero crossing of the supply voltage.		N/A

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
19.11.4.7	The appliance is subjected to mains signals in accordance with IEC 61000-4-13, Table 11 with test level class 2 using the frequency steps according to Table 10.		N/A
19.11.4.8	The appliance is supplied at rated voltage and operated under normal operation.After approximately 60 s, the power supply voltage is reduced to a level such that the appliance ceases to respond to user inputs or parts controlled by the programmable component cease to operate, whichever occurs first.		N/A
	This value of supply voltage is recorded.The appliance is supplied at rated voltage and operated under normal operation. The voltage is then reduced to a value of approximately 10 % less than the recorded voltage. It is held at this value for approximately 60 s and then increased to rated voltage. The rate of decrease and increase of the power supply voltage is to be approximately 10 V/s.		
19.13	During the tests, the appliance shall not emit flames. molten metal, or poisonous or ignitable gas in hazardous amounts and temperature rises shall not exceed the values shown in Table 9.		P
	After the tests. and when the appliance has cooled to approximately room temperature compliance with Clause 8 shall not be impaired and the appliance shall comply with 20.2 if it can still be operated.		P
	When the insulation, other than that of class III appliances or class III constructions that do not contain live parts. has cooled down to approximately room temperature. it shall withstand the electric strength test of 16.3, the test voltage, however, being as specified in Table 4.		P

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
	For appliances which are immersed in or filled with conducting liquid in normal use. the appliance is immersed in or filled with water for 24 h before the electric strength test is carried out.		P
	After the operation or interruption of a control. clearances and creepage distances across the functional insulation shall withstand the electric strength test of 16.3 the test voltage however. being twice the working voltage.		P
	The appliance shall not undergo a dangerous malfunction. and there shall be no failure of protective electronic circuits if the appliance is still operable.		P
	Appliances tested with an electronic switch in the off position, or in the stand-by mode, shall-not become operational, or-if they become operational, not result in a dangerous malfunction during or after the tests of 19.11.4.		P
19.14	Appliances are operated under the conditions of Clause 11. Any contactor or relay contact that operates under the conditions of Clause 11 is short-circuited.		N/A
19.15	For appliances incorporating a mains voltage selector switch, this switch is set to the lowest rated voltage position and the highest value of rated voltage is applied.		N/A
20	STABILITY AND MECHANICAL HAZARDS		/
20.1	Appliances, other than fixed appliances and hand-held appliances, intended to be used on a surface such as the floor or a table, shall have adequate stability.		P
	Compliance is checked by the following test, appliances incorporating an appliance inlet being fitted with an appropriate connector and flexible cord.		P

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
	The appliance, not connected to the supply mains, is placed in any normal position of use on a plane inclined at an angle of 10° to the horizontal, the supply cord resting on the inclined plane in the most unfavourable position. However, if part of an appliance comes into contact , with the horizontal supporting surface when the appliance is tilted through an angle of 10°, the appliance is placed on a horizontal support and tilted in the most unfavourable direction through an angle of 10°.	There is no tilt to the ground	P
	Appliances provided with doors are tested with the doors open or closed. whichever is the more unfavourable.		P
	Appliances intended to be filled with liquid by the user in normal use are tested empty or filled with the most unfavourable quantity of water up to the capacity indicated in the instructions.		P
20.2	Moving parts of appliances shall, as far as is compatible with the use and working of , the appliance, be positioned or enclosed to provide adequate protection against personal injury in normal use.		P
	This requirement does not apply to parts of an appliance that necessarily have to be exposed to allow the appliance to perform its working function.		P
	Protective enclosures, guards and similar parts shall be non-detachable parts and shall have , adequate mechanical strength. However, enclosures that can be opened by overriding an interlock by applying the test probe are considered to be detachable parts.		P

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
	Compliance is checked by inspection, by the test of 21.1 and by applying a force not exceeding 5 n by means of a test probe that is similar to test probe B of IEC 61032 but having a circular stop face with a diameter of 50 mm, instead of the non-circular face.		P
	For appliances provided with movable devices such as those intended for varying the tension of belts. the test with the test probe is carried out with these devices adjusted to the most unfavourable position within their range of adjustment. If necessary. belts are removed.		P
21	MECHANICAL STRENGTH		/
21.1	Appliances shall have adequate mechanical strength and be constructed to withstand.. such rough handling that may be expected in normal use.		P
	Compliance is checked by applying blows to the appliance in accordance with test Ehb of IEC 60068-2-75, the spring hammer test.		P
	The appliance is rigidly supported and three blows, having an impact energy of 0,5 J, are applied to every point of the enclosure that is likely to be weak.		P
	If necessary, the blows are also applied to handles. levers. knobs and similar parts and to signal lamps and their covers but only if the lamps or covers protrude from the enclosure by more than 10 mm or if their surface area exceeds 4 cm2. Lamps within the appliance and their covers are only tested if they are likely to be damaged in normal use.		P

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
	After the test, the appliance shall show no damage that could impair compliance with this , standard and compliance with 8.1. 15.1 and Clause 29 shall not be impaired. In case of doubt supplementary insulation and reinforced insulation are subjected to the electric strength test of 16.3.		P
	Damage to the finish, small dents that do not reduce clearances or creepage distances below the values specified in Clause 29. and small chips that do not adversely affect protection against access to live parts or moisture, are ignored.		P
21.2	Accessible parts of solid insulation shall have sufficient strength to prevent penetration by sharp implements.		P
	Compliance is checked by subjecting the insulation to the following test, unless the thickness . of supplementary insulation is at least 1 mm and that of reinforced insulation is at least 2mm..		P
	The insulation is raised to the temperature measured during the test of Clause 11. The surface of the insulation is then scratched by means of a hardened steel pin, the end of which has the form of a cone with an angle of 40. Its tip is rounded with a radius of0.25 mm ± 0.02 mm. The pin is held at any angle from 80 to 85 to the horizontal and loaded1 so that the force exerted along its axis is 10 n ±0,5 n. The scratches are made by drawing the pin along the surface of the insulation at a speed of approximately 20 mm/s.		P

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
	Two parallel scratches are made. They are spaced sufficiently apart so that they are not affected by each other, their length covering approximately 25 % of the length of the insulation. Two similar scratches are made at 90 to the first pair without crossing them.		P
	The test fingernail of Figure 7 is then applied to the scratched surface with a force of approximately 10 n. No further damage, such as separation of the material, shall occur. The insulation shall then withstand the electric strength test of 16.3.		P
	The hardened steel pin is then applied perpendicularly with a force of 30 n±0.5N to an , unscratched part of the surface. The insulation shall then withstand the electric strength test of 16.3 with the pin still applied and used as one of the electrodes.		P
22	CONSTRUCTION		/
22.2	For stationary appliances, means shall be provided to ensure all-pole disconnection from the supply mains.		P
	Such means shall be one of the following: -- a supply cord fitted with a plug; -- a switch complying with 24.3; -- a statement in the instructions that a disconnection incorporated in the fixed wiring is to be provided; -- an appliance inlet.		P
	Single-pole switches and single-pole protective devices that disconnect heating elements , from the supply mains in single-phase, permanently connected class 0l appliances and class I appliances shall be connected to the phase conductor.		N/A

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
22.3	Appliances with pins for insertion into socket-outlets shall not impose undue strain on these socket-outlets. The means for retaining the pins shall withstand the forces to which the pins are likely to be subjected in normal use.		N/A
	Compliance is checked by inserting the pins of the appliance into a socket-outlet without earthing contact. The socket-outlet has a horizontal pivot at a distance of 8 mm behind the engagement face of the socket-outlet and in the plane of the contact tubes.		N/A
	The torque that has to be applied to maintain the engagement face of the socket-outlet in the vertical plane shall not exceed 0.25 Nm		N/A
	A new sample of the appliance is firmly held so that the retention of the pins is not affected.The appliance is placed in a heating cabinet for 1 h at a temperature of 70℃±2℃. The appliance is then removed from the heating cabinet and a pull force of 50 n is immediately applied for 1 min to each pin along their longitudinal axes.		N/A
	Each pin is then subjected in turn to a torque of 0,4 Nm, which is applied for 1 min in each direction. The pins shall not rotate unless rotation does not impair compliance with this standard		N/A
22.4	Appliances for heating liquids and appliances causing undue vibration shall not be provided with pins for insertion into socket-outlets.		P
22.5	Appliances intended to be connected to the supply mains by means of a plug shall be• constructed so that in normal use there is no risk of electric shock from charged capacitors having a rated capacitance exceeding 0,1 µF, when the pins of the plug are touched.		N/A

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
	The appliance is supplied at rated voltage. Any switch is then placed in the off position and the appliance is disconnected from the supply mains at the instant of voltage peak. One second after disconnection, the voltage between the pins of the plug is measured with an instrument that does not appreciably affect the value to be measured.		P
22.6	Appliances shall be constructed so that their electrical insulation cannot be affected by water that could condense on cold surfaces or by liquid that could leak from containers hoses, couplings and similar parts of the appliance. The electrical insulation of class IL appliances and class II constructions shall not be affected if a hose ruptures or a seal leaks		N/A
	Drops of coloured water solution are applied by a syringe to parts inside the appliance where . leakage of a liquid could occur and affect electrical insulation. The appliance is in operation or at rest, whichever is the more unfavourable.		N/A
22.7	Appliances containing liquid or gases in normal use or having steam-producing devices shall incorporate adequate safeguards against the risk of excessive pressure.		N/A
22.8	For appliances having compartments to which access can be gained without the aid of a tool and that are likely to be cleaned in normal use. the electrical connections shall be arranged so that they are not subject to pulling during cleaning.		P

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
22.9	Appliances shall be constructed so that parts such as insulation. internal wiring, windings. commutators and slip rings are not exposed to oil, grease or similar substances unless the substance has adequate insulating properties so that compliance with the standard is not impaired.		P
22.10	It shall not be possible to reset voltage-maintained non-self-resetting thermal cut outs by the operation of an automatic switching device incorporated within the appliance. This requirement is only applicable if a non-self-resetting thermal cut-out is required by the standard and a voltage maintained non-self-resetting thermal cut-out is used to meet it.		P
	Reset buttons of non-self-resetting controls shall be located or protected so that their accidental resetting is unlikely to occur if this could result in a hazard.		P
22.11	Non-detachable parts that protect against access to live parts. moisture or contact with moving parts shall be fixed in a reliable manner and withstand the mechanical stress occurring during normal use. Snap-in devices used for fixing such parts shall have an obvious locked position. The fixing properties of snap-in devices used in parts that are likely to be removed during installation or servicing shall be reliable.		P
	Parts that are likely to be removed during installation or servicing are disassembled and assembled 10 times before the test is carried out.		P

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
	The test is carried out at room temperature. However, if compliance may be affected by the temperature of the appliance, the test is also carried out immediately after it has been operated under the conditions specified in Clause 11.		P
	The test is applied to all parts that are likely to be detachable whether or not they are fixed by screws. rivets or similar parts.		P
	A force is applied without jerks for 10 s in the most unfavourable direction to parts likely to be weak. The force is as follows: -- push force, 50 n; -- pull force: • if the shape of the part is such that the fingertips cannot easily slip off. 50 n; • if the projection of the part that is gripped is less than 10 mm in the direction of removal. 30 n.		P
	The pull force is applied by a suitable means, such as a suction cup, so that the test results are not affected. While the force is being applied. the test fingernail of Figure 7 is inserted in any aperture or joint with a force of 10 n. The fingernail is then slid sideways with a force of 10 n but is not twisted or used as a lever.		P
22.12	Handles. knobs, grips, levers and similar parts shall be fixed in a reliable manner so that they will not work loose in normal use if loosening could result in a hazard. If these parts are used to indicate the position of switches or similar components, it shall not be possible to fix them incorrectly if this could result in a hazard.		P

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
22.13	Appliances shall be constructed so that when handles are gripped in normal use, contact is unlikely between the operator's hand and parts having a temperature rise exceeding the value specified in Table 3 for handles which are held for short periods only in normal use.		P
22.14	Appliances shall have no ragged or sharp edges, other than those necessary for the . functioning of the appliance, which could create a hazard for the user in normal use or during user maintenance.		P
	Pointed ends of self-tapping screws or other fasteners shall be located so that they are . unlikely to be touched by the user in normal use or during user maintenance.		P
22.15	Storage hooks and similar devices for flexible cords shall be smooth and well rounded.		P
22.16	Automatic cord reels shall be constructed so that they do not cause -- undue abrasion or damage to the sheath of the flexible cord;breakage of conductor strands; -- undue wear of contacts.		N/A
	Two-thirds of the length of the cord is unreeled. If the withdrawable length of the cord is less than 225 cm. the cord is unreeled so that a length of 75 cm remains on the reel. An additional length of 75 cm of the cord is then unreeled and pulled in a direction so that the greatest abrasion occurs to the sheath. taking into account the normal position of use of the appliance.Where the cord leaves the appliance. the angle between the axis of the cord during the test and the axis of the cord when it is unreeled without substantial resistance is approximately 60 The cord is allowed to be recoiled by the reel.		N/A

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
	The test is carried out 6 000 times at a rate of approximately 30 times per minute or at the maximum rate allowed by the construction of the cord reel if this is less.		N/A
22.17	Spacers intended to prevent the appliance from overheating walls shall be fixed so that it is not possible to remove them from the outside of the appliance by hand or by means of a screwdriver or a spanner.		N/A
22.18	Current-carrying parts and other metal parts, the corrosion of which could result in a hazard, shall be resistant to corrosion under normal conditions of use.		P
22.19	Driving belts shall not be relied upon to provide the required level of insulation unless they are constructed to prevent inappropriate replacement.		P
22.20	Direct contact between live parts and thermal insulation shall be effectively prevented unless such material is non-corrosive, non-hygroscopic and non-combustible		P
22.21	Wood. cotton, silk, ordinary paper and similar fibrous or hygroscopic material shall not be used as insulation, unless impregnated. This requirement does not apply to magnesium oxide and mineral ceramic fibres used for the electrical insulation of heating elements.		N/A
22.22	Appliances shall not contain asbestos.		P
22.23	Oils containing polychlorinated biphenyl (PCB) shall not be used in appliances.		P
22.24	Bare heating elements, other than those in class III appliances or class III constructions that do not contain live parts. shall be supported so that the heating conductor is unlikely to come into contact with accessible metal parts if they rupture.		N/A

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
22.25	Appliances shall be constructed so that sagging heating conductors cannot come into . contact with accessible metal parts. This requirement does not apply to class III appliances or class III constructions that do not contain live parts.		N/A
22.26	Appliances having parts of class III construction shall be constructed so that the . insulation between parts operating at safety extra-low voltage and other live parts complies with the requirements for double insulation or reinforced insulation.		N/A
22.27	Parts connected by protective impedance shall be separated by double insulation , or reinforced insulation		N/A
22.28	For class II appliances connected in normal use to the gas mains or to the water mains, metal parts conductively connected to the gas pipes or in contact with the water shall be separated from live parts by double insulation or reinforced insulation.		N/A
22.29	Class II appliances intended to be permanently connected to fixed wiring shall be constructed so that the required degree of access to live parts is maintained after installation.		N/A
22.30	Parts of class II construction which serve as supplementary insulation on reinforced insulation. and which could be omitted during reassembly after servicing, shall be. -- fixed so that they cannot be removed without being seriously damaged,or -- constructed so that they cannot be replaced in an incorrect position and if they are omitted, the appliance is rendered inoperable or manifestly incomplete.		N/A

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
22.31	Neither clearances nor creepage distances over supplementary insulation and reinforced insulation shall be reduced below the values specified in Clause 29 as a result of wear.		P
22.32	Supplementary insulation and reinforced insulation shall be constructed or protected so that the deposition of pollution resulting from wear of parts within the appliance does not reduce clearances or creepage distances below the values specified in Clause 29.		P
	Parts of natural or synthetic rubber used as supplementary insulation shall be resistant to ageing or be located and dimensioned so that creepage distances are not reduced below the values specified in 29.2. even if cracks occur.		P
	Ceramic material which is not tightly sintered. similar materials or beads alone shall not bel used as supplementary insulation or reinforced insulation.		P
	Insulating material in which heating conductors are embedded is considered to be basic insulation and not reinforced insulation. This requirement is not applicable to heating conductors in PTC heating elements.		N/A
22.33	Conductive liquids that are or may become accessible in normal use and conductive liquids that are in contact with unearthed accessible metal parts shall not be in direct contact with live parts. Electrodes shall not be used for heating liquids.		P
	For class II construction, conductive liquids that are or may become accessible in normal e and conductive liquids that are in contact with unearthed accessible metal parts shall be in direct contact with basic insulation or reinforced insulation unless the reinforced insulation consists of at least 3 layers.		N/A

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
	For class II construction, conductive liquids which are in contact with live parts shall not be in direct contact with reinforced insulation unless the reinforced insulation consists of at least 3 layers.		N/A
	An air layer shall not be used as basic insulation or supplementary insulation in a double insulation system if it is likely to be bridged by leaking liquid.		N/A
22.34	Shafts of operating knobs. handles, levers and similar parts shall not be live unless the shaft is inaccessible when the part is removed.		P
22.35	For constructions other than those of class III. handles. levers and knobs that are held or actuated in normal use shall not become live in the event of a failure of basic . insulation. If these handles, levers and knobs are of metal and if their shafts or fixings are likely to become live in the event of a failure of basic insulation. they shall be adequately covered by insulating material or their accessible parts shall be separated from their shafts or fixings by supplementary insulation. (IEC 60335-1)		P
	For stationary appliances, this requirement does not apply to handles, levers and knobs, . other than those of electrical components, provided that they are reliably connected to an earthing terminal or earthing contact or separated from live parts by earthed metal.		P
22.36	For appliances other than those of class III, handles which are continuously held in the hand in normal use shall be constructed so that when gripped in normal use, the operator's hand is not likely to touch metal parts unless they are separated from live parts by double insulation or reinforced insulation.		N/A

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
22.37	For class II appliances, capacitors shall not be connected to accessible metal parts . and their casings, if of metal, shall be separated from accessible metal parts by supplementary insulation..		N/A
	This requirement does not apply to capacitors complying with the requirements for protective impedance specified in 22.42.		N/A
22.38	Capacitors shall not be connected between the contacts of a thermal cut-out.		N/A
22.39	Lampholders shall be used only for the connection of lamps.		N/A
22.40	Motor-operated appliances and combined appliances which are intended to be . moved while in operation, or which have accessible moving parts, shall be fitted with a switch to control the motor. The actuating member of this switch shall be easily visible and accessible.		P
	Unless the appliance can operate continuously, automatically or remotely without giving rise to a hazard, appliances for remote operation shall be fitted with a switch for stopping the, operation of the appliance. The actuating member of this switch shall be easily visible and accessible.		N/A
22.41	Appliances shall not incorporate components, other than lamps, containing mercury.		N/A
22.42	Protective impedance shall consist of at least two separate components. If any one of the components is short-circuited or open-circuited, the values specified in 8.1.4 shall not be exceeded..		N/A
	Component impedances shall be unlikely to change significantly during the lifetime of the appliance.		N/A

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
	Compliance is checked by inspection and by measurement and if necessary, for resistors and, capacitors by the following tests.		N/A
	Resistors are checked by the test of 14.1 a) in IEC 60065 and capacitors are checked by the . tests for class Y capacitors in IEC 60384-14 appropriate to the rated voltage of the appliance.		N/A
22.43	Appliances which can be adjusted for different voltages shall be constructed so that accidental changing of the setting is unlikely to occur.		N/A
22.44	Appliances shall not have an enclosure that is shaped or decorated like a toy. NOTE Examples of such enclosures are those representing animals, characters, persons or scale models.		P
22.45	When air is used as reinforced insulation. the appliance shall be constructed so that . clearances cannot be reduced below the values specified in 29.1.3 due to deformation as al result of an external force applied to the enclosure.		N/A
22.47	Appliances intended to be connected to the water mains shall withstand the water pressure expected in normal use.		N/A
	Compliance is checked by connecting the appliance to a water supply having a static pressure equal to twice the maximum inlet water pressure or 1,2 MPa, whichever is higher, for a period of 5 min.		N/A
22.48	Appliances intended to be connected to the water mains shall be constructed to prevent backsiphonage of non-potable water into the water mains.		N/A

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
22.49	For remote operation, the duration of operation shall be set before the appliance can be started unless the appliance switches off automatically at the end of a cycle or it can operate continuously without giving rise to a hazard.		N/A
22.50	Controls incorporated in the appliance, if any, shall take priority over controls actuated by remote operation.		N/A
22.51	A control on the appliance shall be manually adjusted to the setting for remote operation before the appliance can be operated in this mode. There shall be a visual . indication on the appliance showing that the appliance is adjusted for remote operation. The manual setting and the visual indication of the remote mode are not necessary on appliances that can -- operate continuously, or -- operate automatically, or -- be operated remotely, without giving rise to a hazard		N/A
22.52	Socket-outlets on appliances accessible to the user shall be in accordance with the socket-outlet system used in the country in which the appliance is sold		P
23	INTERNAL WIRING		/
23.1	Wireways shall be smooth and free from sharp edges.		P
	Wires shall be protected so that they do not come into contact with burrs, cooling fins or similar edges which may cause damage to their insulation.		P
	Holes in metal through which insulated wires pass shall have smooth well-rounded surfaces . or be provided with bushings.		P
	Wiring shall be effectively prevented from coming into contact with moving parts.		P

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
23.2	Beads and similar ceramic insulators on live wires shall be fixed or located so that thev cannot change their position or rest on sharp edges. If beads are inside flexible metal conduits, they shall be contained within an insulating sleeve, unless the conduit cannot move in normal use.		P
23.3	Different parts of an appliance that can move relative to each other in normal use or during user maintenance shall not cause undue stress to electrical connections and internal conductors, including those providing earthing continuity. Flexible metallic tubes shall not cause damage to the insulation of the conductors contained within them.		P
	Open-coil springs shall not be used to protect the wiring. If a coiled spring, the turns of which touch one another, is used for this purpose, there shall be an adequate insulating lining in addition to the insulation of the conductors.		P
	The movable part is moved backwards and forwards. so that the conductor is flexed through the largest angle allowed by the construction, the rate of flexing being 30 per minute. The number of flexings is -- 1.0 000, for conductors flexed during normal use; -- 100. for conductors flexed during user maintenance.		P

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
	The appliance shall not be damaged to the extent that compliance with this standard is impaired and it shall be fit for further use. In particular, the wiring and its connections shall withstand the electric strength test of 16.3. the test voltage being reduced to 1 000 V and applied between live parts and accessible metal parts only. In addition, not more than 10 % of the strands of any conductor of the internal wiring between the main part of the appliance and the movable part shall be broken. However. if the wiring supplies circuits that consume no more than 15 W. then no more than 30 % of the strands shall be broken.		P
23.4	Bare internal wiring shall be rigid and fixed so that, in normal use, clearances or creepage distances cannot be reduced below the values specified in Clause 29.		P
23.5	The insulation of internal wiring that is subjected to the supply mains voltage shall withstand the electrical stress likely to occur in normal use.		P
	The basic insulation shall be electrically equivalent to the basic insulation of cords complying with IEC 60227 or IEC 60245 or comply with the following electric strength test.		P
	A voltage of 2 000 V is applied for 15 min between the conductor and metal foil wrapped around the insulation. There shall be no breakdown.		P
23.6	When sleeving is used as supplementary insulation on internal wiring, the sleeving . shall be retained in position by clamping at both ends or be such that it can only be removed by breaking or cutting.		P
23.7	Conductors identified by the colour combination green/yellow shall only be used for earthing conductors.		N/A

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
23.8	Aluminium wires shall not be used for internal wiring.		P
23.9	Stranded conductors shall not be consolidated by soldering where they are subjected to contact pressure, unless the contact pressure is provided by spring terminals.		P
23.10	The insulation and sheath of internal wiring, incorporated in external hoses for the connection of an appliance to the water mains, shall be at least equivalent to that of light , polyvinyl chloride sheathed flexible cord (code designation 60227 IEC 52).		N/A
24	COMPONENTS		/
24.1	Components shall comply with the safety requirements specified in the relevant IEC standards as far as they reasonably apply.		P
	Components that have been previously tested and shown to comply with the resistance to fire requirements in the IEC standard for the relevant component need not be retested provided that -- the severity specified in the component standard is not less than the severity specified in 30.2 of this standard, and -- unless the preselection alternative is used, the test report for the component states whether it complied with the IEC standard for the relevant component with or without flame. Flames existing for a cumulative time not exceeding 2 s during the test are ignored		P

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
	Components that have not been separately tested and found to comply with the relevant IEC standard and components that are not marked or not used in accordance with their marking, . are tested in accordance with the conditions occurring in the appliance, the number of samples being that required by the relevant standard.		P
	Lampholders and starterholders that have not been previously tested and found to comply with the relevant IEC standard are tested as a part of the appliance and shall additionally comply with the gauging and interchangeability requirements of the relevant IEC standard under the conditions occurring in the appliance. Where the relevant IEC standard specifies these gauging and interchangeability requirements at elevated temperatures, the temperatures measured during the tests of Clause 11 are used.		P
	There are no additional tests specified for nationally standardized plugs such as those detailed in IEC/TR 60083 or connectors complying with the standard sheets of IEC 60320-1u and IEC 60309.		P
24.1.1	The relevant standard for capacitors likely to be permanently subjected to the supply voltage and used for radio interference suppression or for voltage dividing is IEC 60384-14.		N/A

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
	Capacitors likely to be permanently subjected to the supply voltage are capacitors incorporated in appliances; -- for which 30.2.3 is applicable; or -- for which 30.2.2 is applicable, unless the capacitor is disconnected from the supply mains by an on-off switch. This switch shall provide all-pole disconnection if the capacitor is connected to earth.		N/A
24.1.2	The relevant standard for safety isolating transformers is IEC 61558-2-6. If they , have to be tested, they are tested in accordance with Annex G.		N/A
24.1.5	The relevant standard for appliance couplers is IEC 60320-1. However, for appliances, classified higher than IPXo, the relevant standard is IEC 60320-2-3.		N/A
24.1.6	The relevant standard for small lampholders similar to E10 lampholders is IEC 60238.. the requirements for E10 lampholders being applicable. However, they need not accept a lamp with an E10 cap complying with the current edition of standard sheet 7004-22 of IEC 60061-1.		N/A
24.1.7	If the remote operation of the appliance is via a telecommunication network, the relevant standard for the telecommunication interface circuitry in the appliance is IEC 62151.		N/A
24.1.8	The relevant standard for thermal links is IEC 60691. Thermal links that do not comply with IEC 60691 are considered to be an intentionally weak part for the purposes of Clause 19.		N/A

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
24.1.9	Contactors and relays, other than motor starting relays, are tested as part of the appliance. However, they are also tested in accordance with Clause 17 of IEC 60730-1 under the maximum load conditions occurring in the appliance for at least the number of cycles of operation in 24.1.4 selected according to the contactor or relay function in the appliance.		N/A
24.2	Appliances shall not be fitted with -- switches or automatic controls in flexible cords; -- devices that cause the protective device in the fixed wiring to operate in the event of a fault in the appliance; -- thermal cut-outs that can be reset by a soldering operation, unless the solder has a melting point of at least 230℃.		P
24.3	Switches intended to ensure all-pole disconnection of stationary appliances, as required in 22.2, shall be directly connected to the supply terminals and shall have a contact separation in all poles, providing full disconnection under overvoltage category lll conditions.		P
24.4	Plugs and socket-outlets for extra-low voltage circuits, and those used as terminal .devices for heating elements, shall not be interchangeable with plugs and socket-outlets listed in IEC/TR 60083 or IEC 60906-1 or with connectors and appliance inlets complying with the standard sheets of IEC 60320-1.		N/A
24.5	Capacitors in auxiliary windings of motors shall be marked with their rated voltage and their rated capacitance and shall be used in accordance with these markings.		N/A

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
24.6	The working voltage of motors directly connected to the supply mains and having basic insulation that is inadequate for the rated voltage of the appliance, shall not exceed 42 V. In addition, they shall comply with the requirements of Annex I.		N/A
24.7	Detachable hose-sets for the connection of appliances to the water mains shall comply with IEC 61770. They shall be supplied with the appliance.		N/A
	Appliances intended to be permanently connected to the water mains shall not be connected y a detachable hose-set.		N/A
24.8	Motor running capacitors in appliances for which 30.2.3 is applicable and that are permanently connected in series with a motor winding shall not cause a hazard in the event of a capacitor failure.		N/A
	The requirement is considered to be met by one or more of the following conditions; -- the capacitors are of class of safety protection P2 according to IEC 60252-1; -- the capacitors are housed within a metallic or ceramic enclosure that will prevent the emission of flame or molten material resulting from failure of the capacitor; NOTE: The enclosure may have an entry or exit hole for the wiring connecting the capacitor to the motor. -- the distance of separation of the outer surface of the capacitor to adjacent non-metallic parts exceeds 50 mm;		N/A

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
	-- adjacent non-metallic parts within 50 mm of the outer surface of the capacitor withstand the needle-flame test of Annex E; -- adjacent non-metallic parts within 50 mm of the outer surface of the capacitor are classified as at least V-1 according to IEC 60695-11-10. provided that the test sample used for the classification was no thicker than the relevant part of the appliance.		N/A
25	SUPPLY CONNECTION AND EXTERNAL FLEXIBLE CORDS		/
25.2	Appliances, other than stationary appliances for multiple supply, shall not be provided with more than one means of connection to the supply mains. Stationary appliances for multiple supply may be provided with more than one means of connection provided that the relevant circuits are adequately insulated from each other.		N/A
25.3	Appliances intended to be permanently connected to fixed wiring shall be provided with one of the following means for connection to the supply mains: -- a set of terminals allowing the connection of a flexible cord; NOTE In this case, a cord anchorage must also be provided. -- a fitted supply cord; -- a set of supply leads accommodated in a suitable compartment; -- a set of terminals allowing the connection of cables of fixed wiring having the nominal cross-sectional areas specified in 26.6;		N/A
	-- a set of terminals and cable entries, conduit entries, knock-outs or alands. which allow the connection of the appropriate types of cable or conduit.		N/A

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
	Appliances intended to be permanently connected to fixed wiring that are provided with • a set of terminals allowing the connection of cables of fixed wiring having the nominal cross-sectional areas specified in 26.6, or • a set of terminals and cable entries. conduit entries, knock-outs or glands, which allow the connection of the appropriate types of cable or conduit, shall allow the connection of the supply conductors after the appliance has been fixed to its support.		N/A
25.4	For appliances intended to be permanently connected to the fixed wiring and having a . rated current not exceeding 16 A, cable and conduit entries shall be suitable for cables or conduits having a maximum overall dimension shown in Table 10		N/A
	Conduit entries, cable entries and knock-outs shall be constructed or located so that the . introduction of the conduit or cable does not reduce clearances or creepage distances below the values specified in Clause 29.		N/A
25.8	Conductors of supply cords shall have a nominal cross-sectional area not less than that shown in Table 11.		P
25.9	Supply cords shall not be in contact with sharp points or edges of the appliance.		P
25.10	The supply cord of class I appliances shall have a green/yellow core that is connected to the earthing terminal of the appliance and for appliances not intended for permanent connection to the fixed wiring, to the earthing contact of the plug.		P

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
25.11	Conductors of supply cords shall not be consolidated by soldering where they are subjected to contact pressure, unless the contact pressure is provided by spring terminals.		P
25.12	The insulation of the supply cords shall not be damaged when moulding the cord to part of the enclosure.		P
25.13	Inlet openings for supply cords shall be constructed so that the sheath of the supply cord can be introduced without risk of damage. Unless the enclosure at the inlet opening is insulating material, a non-detachable lining or non-detachable bushing shall be provided that complies with 29.3 for supplementary insulation. If the supply cord is unsheathed, a similar additional bushing or lining is required, unless the appliance is a class 0 appliance or a class III appliance that does not contain live parts		N/A
25.14	Appliances provided with a supply cord and that are moved while in operation shall . be constructed so that the supply cord is adequately protected against excessive flexing where it enters the appliance		N/A
	The part of the appliance that includes the inlet opening is fixed to the oscillating member so . that. when the supply cord is at the middle of its travel. the axis of the cord where it enters the cord quard or inlet is vertical and passes through the axis of oscillation. The major axis of the section of flat cords shall be parallel to the axis of oscillation.		N/A
	The cord is loaded so that the force applied is -- 10 n, for cords having a nominal cross-sectional area exceeding 0,75 mm^2; -- 5N, for other cords.		N/A

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
	The distance X, as shown in Figure 8, between the axis of oscillation and the point where the cord or cord guard enters the appliance, is adjusted so that when the oscillating member moves over its full range. the cord and load make the minimum lateral movement.		N/A
	The oscillating member is moved through an angle of 90 (45 on either side of the vertical), the number of flexings for type Z attachments being 20 000 and for other attachments.10000. The rate of flexing is 60 per minute.		N/A
25.15	Appliances provided with a supply cord, and appliances intended to be permanently connected to fixed wiring by a flexible cord. shall have a cord anchorage. The cord anchorage."shall relieve conductors from strain, including twisting, at the terminals and protect the insulation of the conductors from abrasion.		P
	A mark is made on the cord while it is subjected to the pull force shown in Table 12, at a distance of approximately 20 mm from the cord anchorage or other suitable point.		P
	The cord is then pulled, without jerking. for 1 s in the most unfavourable direction with the force specified. The test is carried out 25 times. The cord. unless on an automatic cord reel, is then subjected to a torque that is applied as close as possible to the appliance. The torque is specified in Table 12 and is applied for 1min.		P

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
25.16	Cord anchorages for type X attachments shall be constructed and located so that -- replacement of the cord is easily possible; -- it is clear how the relief from strain and the prevention of twisting are obtained; -- they are suitable for the different types of supply cord that may be connected, unless the cord is specially prepared; -- the cord cannot touch the clamping screws of the cord anchorage if these screws are accessible, unless they are separated from accessible metal parts by supplementary insulation; -- the cord is not clamped by a metal screw which bears directly on the cord;		N/A
25.17	For type Y attachment and type Z attachment, cord anchorages shall be adequate. ,		P
25.18	Cord anchorages shall be arranged so that they are only accessible with the aid of a tool or shall be constructed so that the cord can only be fitted with the aid of a tool.		P
25.19	For type X attachment, glands shall not be used as cord anchorages in portable appliances. Tying the cord into a knot or tying the cord with string is not allowed.		N/A
25.20	The insulated conductors of the supply cord for type Y attachment and type Z attachment shall be additionally insulated from accessible metal parts by basic insulation for class 0 appliances, class 0I appliances and class I appliances, and by supplementary insulation for class II appliances. This insulation may be provided by the sheath of the supply cord or by other means.		P

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
25.21	The space for the connection of supply cords having type X attachment, or for the connection of fixed wiring. shall be constructed -- so that it is possible to check that the supply conductors are correctly positioned and connected before fitting any cover; -- so that any cover can be fitted without risk of damage to the conductors or their insulation; -- for portable appliances, so that the uninsulated end of a conductor, should it become free from the terminal, cannot come into contact with accessible metal parts.		N/A
	Portable appliances are subjected to the following additional test unless they are provided with pillar terminals and the supply cord is clamped within 30 mm of them.		N/A
	The clamping screws or nuts are loosened in turn. A force of 2 n is applied to the conductor in any direction at a position adjacent to the terminal. The uninsulated end of the conductor shall not come into contact with accessible metal parts.		N/A
25.22	Appliance inlets shall		N/A

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
	-- be located or enclosed so that live parts are not accessible during insertion or removal of the connector. This requirement is not applicable to appliance inlets complying with IEC 60320-1; -- be located so that the connector can b inserted without difficulty; -- be located so that. after insertion of the connector, the appliance is not supported by the connector when it is placed in any position of normal use on a flat surface; -- not be an appliance inlet for cold conditions if the temperature rise of external metal parts of the appliance exceeds 75 K during the test of Clause 11. unless the supply cord is unlikely to touch such metal parts in normal use.		N/A
25.23	Interconnection cords shall comply with the requirements for the supply cord, , except that -- the cross-sectional area of the conductors of the interconnection cord is determined on the basis of the maximum current carried by the conductor during the test of Clause 11 and not by the rated current of the appliance; -- the thickness of the insulation of the conductor may be reduced if the voltage of the conductor is less than the rated voltage.		P
25.24	Interconnection cords shall not be detachable without the aid of a tool if compliance, with this standard is impaired when they are disconnected.		N/A
25.25	The dimensions of pins of appliances that are inserted into socket-outlets shall be compatible with the dimensions of the relevant socket-outlet. Dimensions of the pins and engagement face are to be in accordance with the dimensions of the relevant plug listed in IECITR 60083.		N/A

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
26	TERMINALS FOR EXTERNAL CONDUCTORS		/
26.1	Appliances shall be provided with terminals or equally effective devices for the connection of external conductors. The terminals, other than terminals in class III appliances that do not contain live parts, shall only be accessible after the removal of a non-detachable" cover. However, earthing terminals may be accessible if a tool is required to make the connections and means are provided to clamp the wire independently from its connection.		N/A
26.2	Appliances having type X attachment, except those having a specially prepared cord, and appliances for the connection of cables of fixed wiring shall be provided with terminals inn which the connections are made by means of screws. nuts or similar devices. unless the connections are soldered..		N/A
	The screws and nuts shall not be used to fix any other component except that they may also . clamp internal conductors if these are arranged so that they are unlikely to be displaced when fitting the supply conductors.		N/A
26.3	Terminals for type X attachment and those for the connection of cables of fixed wiring shall be constructed so that they clamp the conductor between metal surfaces with sufficient contact pressure but without causing damage to the conductor.		N/A

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
	The terminals shall be fixed so that when the clamping means is tightened or loosened -- the terminal does not become loose. This does not apply if the terminals are fixed with two screws, or are fixed with one screw in a recess so that there is no appreciable movement or if they are not subject to torsion in normal use and they are locked by a self-hardening resin; -- internal wiring is not subjected to stress; -- neither clearances nor creepage distances are reduced below the values specified in Clause 29		N/A
26.4	Terminals for type X attachment, except type X attachments having a specially . prepared cord, and terminals for the connection of cables of fixed wiring, shall not require special preparation of the conductor such as by soldering of the strands of the conductor, the use of cable lugs, eyelets or similar devices. They shall be constructed or placed so that the conductor cannot slip out when clamping screws or nuts are tightened.		N/A
26.5	Terminals for type X attachment shall be located or shielded so that if a wire of a . stranded conductor escapes when the conductors are fitted, there is no risk of accidental connection to other parts that could result in a hazard.		N/A
	A 8 mm length of insulation is removed from the end of a flexible conductor having a nominal cross-sectional area as specified in Table 11. One wire of the stranded conductor is left free and the other wires are fully inserted and clamped in the terminal. The free wire is bent without tearing the insulation back, in every possible direction but without making sharp bends around barriers.		N/A

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
26.6	Terminals for type X attachment and for the connection of cables of fixed wiring shall . allow the connection of conductors having the nominal cross-sectional areas shown in.Table 13. However, if a specially prepared cord is used, the terminals need only be suitable for the connection of that cord.		N/A
26.7	Terminals for type X attachment, other than those in class III appliances that do not contain live parts, shall be accessible after removal of a cover or part of the enclosure.		N/A
26.8	Terminals for the connection of fixed wiring, including the earthing terminal, shall be located close to each other.		N/A
26.9	Terminals of the pillar type shall be constructed and located so that the end of a . conductor introduced into the hole is visible, or can pass beyond the threaded hole for a distance egual to half the nominal diameter of the screw but at least 2.5 mm.		N/A
26.10	Terminals with screw clamping and screwless terminals shall not be used for the connection of the conductors of flat twin tinsel cords unless the ends of the conductors are fitted with means suitable for use with screw terminals.		N/A

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
26.11	For appliances having type Y attachment or type Z attachment, soldered, welded crimped or similar connections may be used for the connection of external conductors. For class II appliances, the conductor shall be positioned or fixed so that reliance is not placed . upon the soldering, crimping or welding alone to maintain the conductor in position. However, these methods may be used alone if barriers are provided so that clearances and creepage distances between live parts and other metal parts cannot be reduced below the values specified for supplementary insulation, if the conductor becomes free at the soldered or welded joint or slips out of the crimped connection.		N/A
27	PROVISION FOR EARTHING		/
27.1	Accessible metal parts of Class 0I and I appliances, permanently and reliably connected to an earthing terminal or contact of the appliance inlet		P
	Earthing terminals not connected to neutral terminal		P
	Class 0, II and III appliance have no provision for earthing		N/A
	Safety extra-low voltage circuits not earthed, unless protective extra-low voltage circuits		N/A
27.2	Clamping means adequately secured against accidental loosening		P
	Terminals used for the connection of external equipotential bonding conductors allow connection of conductors of 2.5 to 6 mm², and		N/A
	do not provide earthing continuity between different parts of the appliance		P
	Conductors cannot be loosened without the aid of a tool		P

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
27.3	For appliances with supply cord, current-carrying conductors become taut before earthing conductor, if the cord slips out of the cord anchorage		N/A
A	If a detachable part having an earth connection is plugged into another part of the appliance, the earth connection shall be made before the current-carrying connections are established. (IEC 60335-1:01 + A1:2004)		P
A	The current-carrying connections shall be separated before the earth connection when removing the part (IEC 60335-1:01 + A1:2004)		P
27.4	No risk of corrosion resulting from contact between metal of earthing terminal and other metal		P
	Adequate resistance to corrosion of coated or uncoated parts providing earthing continuity, other than parts of a metal frame or enclosure		P
	Parts of steel providing earthing continuity provided at the essential areas with an electroplated coating, thickness at least 5 µm		P
	Adequate protection against rusting of parts of coated or uncoated steel, only intended to provide or transmit contact pressure		P
	In case of aluminium alloys precautions taken to avoid risk of corrosion		N/A
27.5	Low resistance of connection between earthing terminal and earthed metal parts		P
	This requirement does not apply to connections providing earthing continuity in the protective extra- low voltage circuit, provided that clearances of basic insulation are based on the rated voltage of the appliance		N/A
	Resistance not exceeding 0,1 Ω at the specified low-resistance test	0.027Ω	P

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
27.6	The printed conductors of printed circuit boards not used to provide earthing continuity in hand held appliances		N/A
	They may be used in other appliances if:		P
	- at least two tracks are used with independent soldering points and the appliance complies with requirements of 27.5 for each circuit		N/A
28	SCREWS AND CONNECTIONS		/
28.1	Fixings. the failure of which may impair compliance with this standard, electrical . connections and connections providing earthing continuity shall withstand the mechanical, stresses occurring in normal use.		P
	Screws used for these purposes shall not be of metal that is soft or liable to creep, such as . zinc or aluminium. If they are of insulating material, they shall have a nominal diameter of at least 3 mm and they shall not be used for any electrical connections or connections providing earthing continuity.		P
	Screws used for electrical connections or for connections providing earthing continuity shall screw into metal		P
	Screws shall not be of insulating material if their replacement by a metal screw could impair supplementary insulation or reinforced insulation. Screws that may be removed when• replacing a supply cord having a type X attachment or when undertaking user maintenance shall not be of insulating material if their replacement by a metal screw could impair basic insulation.		P

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
28.2	Electrical connections and connections providing earthing continuity shall be . constructed so that contact pressure is not transmitted through non-ceramic insulating material that is liable to shrink or to distort unless there is sufficient resiliency in the metallic parts to compensate for any possible shrinkage or distortion of the insulating material.		N/A
	This requirement does not apply to electrical connections in circuits of appliances for which. • 30.2.2 is applicable and that carry a current not exceeding 0,5 A; • 30.2.3 is applicable and that carry a current not exceeding 0,2 A		N/A
28.3	Space-threaded (sheet metal) screws shall only be used for electrical connections if they clamp the parts together.		N/A
	Thread-cutting (self-tapping) screws and thread rolling screws shall only be used for electrical connections if they generate a full form standard machine screw thread. However, thread cutting (self-tapping) screws shall not be used if they are likely to be operated by the user or installer.		N/A
	Thread-cutting. thread rolling and space-threaded screws may be used in connections providing earthing continuity provided it is not necessary to disturb the connection -- in normal use		N/A
	-- during user maintenance, -- when replacing a supply cord having a type X attachment, or -- during installation.		N/A

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
	At least two screws must be used for each connection providing earthing continuity, unless the screw forms a thread having a length of at least half the diameter of the screw.		N/A
28.4	Screws and nuts that make a mechanical connection between different parts of the appliance shall be secured against loosening if they also make electrical connections or connections providing earthing continuity.		
	This requirement does not apply to screws in the earthing circuit if at least two screws are used for the connection or if an alternative earthing circuit is provided.		N/A
	Rivets used for electrical connections or for connections providing earthing continuity shall be secured against loosening if these connections are subject to torsion in normal use.		N/A
29	CLEARANCES CREEPAGE DISTANCES AND SOLID INSULATION		/
	Appliances shall be constructed so that the clearances, creepage distances and solid insulation are adequate to withstand the electrical stresses to which the appliance is liable to,• be subjected.		P
29.1	Clearances shall not be less than the values specified in Table 16, taking into account the rated impulse voltage for the overvoltage categories of Table 15, unless. for basic insulation and functional insulation, they comply with the impulse voltage test of Clause 14.. However, if the construction is such that the distances could be affected by wear, by distortion, by movement of the parts or during assembly, the clearances for rated impulse voltages of 1 500 V and above are increased by 0,5 mm and the impulse voltage test is not applicable.		P

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
	The impulse voltage test is not applicable when the microenvironment is pollution degree 3 or for basic insulation of class 0 appliances and class 0l appliances.		P
29.1.1	The clearances of basic insulation shall be sufficient to withstand the overvoltages likely to occur during use, taking into account the rated impulse voltage. The values of Table 16, or the impulse voltage test of Clause 14, are applicable.		P
	The clearance at the terminals of tubular sheathed heating elements may be reduced to1.0 mm if the microenvironment is pollution degree 1.		N/A
	Lacquered conductors of windings are considered to be bare conductors.		P
29.1.2	Clearances of supplementary insulation shall be not less than those specified for basic insulation in Table 16.		P
29.1.3	Clearances of reinforced insulation shall be not less than those specified for basic insulation in Table 16. using the next higher step for rated impulse voltage as a reference.		P
	Compliance is checked by measurement. For double insulation, when there is no intermediate conductive part between the basic insulation and supplementary insulation, clearances are measured between live parts and the accessible surface, and the insulation system is treated as reinforced insulation as shown in Figure 11.		P

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
29.1.4	The clearances for functional insulation are the largest values determined from -- Table 16 based on the rated impulse voltage; -- Table F.7a in IEC 60664-1 based on the steady-state voltage or recurring peak voltage expected to occur across it, if the frequency of the steady-state voltage or recurring peak voltage does not exceeds 30 kHz; -- Clause 4 of IEC 60664-4 based on the steady-state voltage or recurring peak voltage expected to occur across it, if the frequency of the steady-state voltage or recurring peak voltage exceeds 30 kHz.		P
29.1.5	For appliances having higher working voltages than rated voltage, for example on . the secondary side of a step-up transformer, or if there is a resonant voltage, the clearances for basic insulation are the largest values determined from -- Table 16 based on the rated impulse voltage; -- Table F.7a in IEC 60664-1 based on the steady-state voltage or recurring peak voltage expected to occur across it, if the frequency of the steady-state voltage or recurring peak voltage does not exceeds 30 kHz; -- Clause 4 of IEC 60664-4 based on the steady-state voltage or recurring peak voltage expected to occur across it, if the frequency of the steady-state voltage or recurring peak voltage exceeds 30 kHz.		N/A

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
	Pollution degree 2 applies unless -- precautions have been taken to protect the insulation, in which case pollution degree 1 applies; -- the insulation is subjected to conductive pollution, in which case pollution degree 3 applies.		P
	In a double insulation system, the working voltage for both the basic insulation and supplementary insulation is taken as the working voltage across the complete double insulation system. It is not divided according to thickness and dielectric constants of the basic insulation and supplementary insulation.		P
29.2.1	Creepage distances of basic insulation shall not be less than those specified in Table 17. However, if the working voltage is periodic and has a frequency that exceeds 30 kHz. the creepage distances shall also be determined from Table 2 of IEC 60664-4.These values shall be used instead if they exceed the values in Table 17.		P
	Except for pollution degree 1, if the test of Clause 14 has been used to check a particular clearance, the corresponding creepage distance shall not be less than the minimum dimension specified for the clearance of Table 16.		P
29.2.2	Creepage distances of supplementary insulation shall be at least those specified for basic insulation in Table 17 or Table 2 of IEC 60664-4, as applicable.		N/A
29.2.3	Creepage distances of reinforced insulation shall be at least double those specified for basic insulation in Table 17 or Table 2 of IEC 60664-4, as applicable.		P

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
29.2.4	Creepage distances of functional insulation shall be not less than those specified in Table 18. However, if the working voltage is periodic and has a frequency that exceeds 30 kHz, the creepage distances shall also be determined from Table 2 of IEC 60664-4.These values shall be used instead if they exceed the values in Table 18.		P
29.3	Supplementary insulation and reinforced insulation shall have adequate thickness, or have a sufficient number of layers, to withstand the electrical stresses that can be expected during the use of the appliance.		P
	Compliance is checked -- by measurement. in accordance with 29.3.1. or -- by an electric strength test in accordance with 29.3.2. if the insulation consists of more than one separate layer, other than natural mica or similar flaky material, or		P
	-- by an assessment of the thermal quality of the material combined with an electric strength test. in accordance with 29.3.3 and for accessible parts of reinforced insulation consisting of a single layer, by measurement in accordance with 29.3.4, or --as specified in Subclause 6.3 of IEC 60664-4 for insulation that is subjected to any periodic voltage having a frequency that exceeds 30 kHz.		P
29.3.1	The thickness of the insulation shall be at least -- 1 mm for supplementary insulation; -- 2 mm for reinforced insulation.		N/A

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
29.3.2	Each layer of material shall withstand the electric strength test of 16.3 for supplementary insulation. Supplementary insulation shall consist of at least 2 layers of material and reinforced insulation of at least 3 layers.		N/A
29.3.3	The insulation is subjected to the dry heat test Bb of IEC 60068-2-2 for 48 h at a temperature of 50 K in excess of the maximum temperature rise measured during the test of Clause 19. At the end of the period, the insulation is subjected to the electric strength test of16.3 at the conditioning temperature and also after it has cooled down to room temperature.		N/A
29.3.4	The thickness of the accessible parts of reinforced insulation consisting of a single layer shall not be less than those specified in Table 19.		N/A
30	RESISTANCE TO HEAT AND FIRE		/
30.1	External parts of non-metallic material, parts of insulating material supporting live parts including connections, and parts of thermoplastic material providing supplementary insulation or reinforced insulation, shall be sufficiently resistant to heat if their deterioration could cause the appliance to fail to comply with this standard.		P
	The test is carried out at a temperature of 40℃±2℃ plus the maximum temperature rise determined during the test of Clause 11. but it shall be at least -- 75℃±2℃, for external parts; -- 125℃±2℃, for parts supporting live parts.		P
30.2	Parts of non-metallic material shall be resistant to ignition and spread of fire. (IEC 60335-1)		P

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
	The requirement does not apply to parts having a mass not exceeding 0,5 g which are considered insignificant parts, provided the cumulative effect of insignificant parts located within 3 mm of each other is unlikely to propagate flames that originate inside the appliance by propagating flames from one insignificant part to another.		P
	The requirement also does not apply to decorative trims, knobs and other parts unlikely to be ignited or to propagate flames that originate inside the appliance.		P
	Appliances for remote operation are considered to be appliances that are operated while unattended and consequently they are subjected to the test of 30.2.3		P
	The tests are carried out on parts of non-metallic material that have been removed from the appliance. When the glow-wire test is carried out, the parts are placed in the same orientation as they would be in normal use.		P
30.2.1	The glow-wire test is also not carried out on parts of material classified at least HB40 . according to IEC 60695-11-10 provided that the test sample used for the classification was no thicker than the relevant part of the appliance.		P
	Parts for which the glow-wire test cannot be carried out, such as those made of soft or foamy material, shall meet the requirements specified in /SO 9772 for material classified HBF, the test sample used for the classification being no thicker than the relevant part of the appliance.		P

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
30.2.3	Appliances that are operated while unattended are tested as specified in 30.2.3.1 and 30.2.3.2. However, the tests are not applicable to: -- parts supporting welded connections and parts within a distance of 3 mm of these connections; -- parts supporting connections in low-power circuits described in 19.11.1 and parts within a distance of 3 mm of these connections; -- soldered connections on printed circuit boards and parts within a distance of 3 mm of these connections; -- connections on small components on printed circuit boards, such as diodes, transistors resistors, inductors, integrated circuits and capacitors not directly connected to the supply mains, and parts within a distance of 3 mm of these connections.		P
30.2.3.1	Parts of non-metallic material supporting connections that carry a current exceeding 0,2 A during normal operation, and parts of non-metallic material, other than small parts, within a distance of 3 mm of such connections, are subjected to the glow-wire test of IEC 60695-2-11 with a test severity of 850 C.		P
	Where a non-metallic material is within a distance of 3 mm of a current-carrying connection, but is shielded from the connection by a different material, the glow-wire test of IEC 60695-2-11 is carried out at the relevant temperature with the tip of the glow-wire applied to the interposed shielding material with the shielded material in place and not directly to the shielded material.		P

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
30.2.3.2	Parts of non-metallic material supporting connections and parts of non-metallic . material within a distance of 3 mm of such connections are subjected to the glow-wire test of IEC 60695-2-11.		P
	The test severity is --750℃, for connections that carry a current exceeding 0,2 A during normal operation, -- 650℃, for other connections.		P
	Where a non-metallic material is within a distance of 3 mm of a current-carrying connection but is shielded from the connection by a different material. the glow-wire test of IEC 60695-2-11 is carried out at the relevant test severity with the tip of the glow-wire applied to the interposed shielding material with the shielded material in place and not directly to the shielded material.		P
30.2.4	The base material of printed circuit boards is subjected to the needle-flame test (NFT) of Annex E. The flame is applied to the edge of the board where the heat sink effect is lowest when the board is positioned as in normal use.		P

US IEC 60335-1:2010;US IEC 60335-2-7:2012			
Clause	Requirement + Test	Result - Remark	Verdict
	The needle-flame test (NFT) of Annex E is not carried out -- on printed circuit boards of low-power circuits described in 19.11.1; -- on the printed circuit boards in • a metal enclosure that confines flames or burning droplets • hand-held appliances, • appliances that have to be kept switched on by hand or foot • appliances that are continuously loaded by hand, -- on a base material classified as V-0 according to IEC 60695-11-10 or VTM-0 according to ISO 9773. provided that the test sample used for the classification was no thicker than the printed circuit board.		N/A
31	RESISTANCE TO RUSTING		/
	Ferrous parts, the rusting of which might cause the appliance to fail to comply with this standard. shall be adequately protected against rusting.		P
32	RADIATION, TOXICITY AND SIMILAR HAZARDS		/
	Appliances shall not emit harmful radiation or present a toxic or similar hazard due to their operation in normal use.		P
	Compliance is checked by the limits or tests specified in part 2. However, if no limits or tests are specified in part 2, then the appliance is deemed to comply with the requirement without . testing.		P

10.1	TABLE: Power input deviation				P
Input deviation of/at:	P rated (W)	P measured (W)	dP (W, %)	Required dP (W, %)	Remark
240V, 50Hz	2500	2580	+6.7%	+20 %	P
240V, 50Hz	2500	2580	+7.3%	+20 %	P
240V, 50Hz	2500	2580	+10.0%	+20%	P
Supplementary information: None					

10.2	TABLE: Current deviation				N/A
Current deviation of/at:	I rated (A)	I measured (A)	dI (A, %)	Required dI (A, %)	Remark
Supplementary information:					

11.8	TABLE: Heating test, thermocouples		N/A
	Test voltage (V) ..... :	1.06×240V=254.4V	–
	Ambient (°C)..... :	45	———

Thermocouple locations	dT (K)	Max. dT (K)
Separation of power cord	16.7	30
Power cord	15.2	30
Ambient of button	15.3	30
Button	5.3	For cl.30
Interlock switch	20.9	30
Carbon brush	27.4	30
Winding	97.8	120
Internal wire	61.9	35
Enclosure inside	15.5	For cl.30
Test floor	8.6	45

13.2	TABLE: Leakage current		P
	Heating appliances: 1.15 x rated input ..... :	1.15×240V	—
	Motor-operated and combined appliances:	---	—
	1.06 x rated voltage..... :		

Leakage current between	I (mA)	Max. allowed I (mA)
Live parts to enclsoure	0.23	0.75
	0.02	0.25

14	TABLE: Transient overvoltages				N/A
Clearance between:	Cl (mm)	Required Cl (mm)	Rated impulse voltage (V)	Impulse test voltage (V)	Flashover (Yes/No)
Supplementary information:					

16.2	TABLE: Leakage current		P
	Single phase appliances: 1.06 x rated voltage .. :	1.06×240V=254.4V	—
	Three phase appliances 1.06 x rated voltage divided by √3:..... :	--	—
Leakage current between		I (mA)	Max. allowed I (mA)
Class Ⅰ Between live part and metal Enclosure		0.23	≦0.75

16.3	TABLE: Electric strength		P
Test voltage applied between:		Voltage (V)	Breakdown (Yes/No)
Input to metal enclosure		1250	No
Input to Knob		3000	No

17	TABLE: Overload protection, temperature rise				N/A
Temperature rise of part/at:			dT (K)	Max. dT (K)	

19.7	TABLE: Abnormal operation, locked rotor/moving parts				N/A
	Test voltage (V) .....:				—
	Ambient, t_1 (°C).....:				—
	Ambient, t_2 (°C).....:				—
Temperature of winding	R_1 (Ω)	R_2 (Ω)	dT (K)	T (°C)	Max. T (°C)
	--	--	--	--	--

24.1	TABLE: Components				P
Object / part No.	Manufacturer/ trademark	Type / model	Technical data	Standard	Mark(s) of conformity
Plug	Various	Various	220V~, 8.3A	---	BS or other EU mark
Power Supply Cord	Various	Various	H05VV-F, 3×0.75 mm2	---	VDE
Knob	Various	Various	V-0, 115℃	---	UL

Switch	Various	Various	8.3A, 220-250V~	---	Tested with appliance
Internal Wire	Various	Various	22AWG, 250℃	---	VDE
Sleeving	Various	Various	V-W, 200℃	---	VDE
Clip	Various	Various	V-2, 115℃	---	VDE
[1)] An asterisk indicates a mark which assures the agreed level of surveillance					

Threaded part identification	Diameter of thread (mm)	Column number (I, II, or III)	Applied torque (Nm)
Earthing terminal	3.87	II	1.2
Enclosure	3.92	II	1.2

29.1	**TABLE: Clearances**					P
	Overvoltage category.. :	II				—
		Type of insulation:				
Rated impulse voltage (V):	Min. cl (mm)	Basic	Functional	Supplementary	Reinforced	Verdict / Remark
	0,5*	--	--	--		--
	0,5*	--	--	--		--
	0,5*	--	--	--		--
	0,5*/**	--	--	--		--
	1,5**	> 2	--	> 2	> 3	P
	3,0**	--	--	--	--	--
	5,5**	--	--	--	--	--
	8,0**	--	--	--	--	--
	11,0**	--	--	--	--	--
*) The value is increased to 0,8mm for pollution degree 3 *) If the construction is affected by wear, distortion, movement of the parts or during assembly, the value is increased by 0,5 mm						

29.2	**TABLE: Creepage distances, basic, supplementary and reinforced insulation**	P
Working voltage (V)	Creepage distance (mm) Pollution degree	

	1	2			3			Type of insulation			
		Material group			Material group			II			
		I	II	IIIa/IIIb	I	II	IIIa/IIIb	B*)	S*)	R*)	Verdict

≤50	0,2	0,6	0,9	1,2	1,5	1,7	1,9	—	—	—	
≤50	0,2	0,6	0,9	1,2	1,5	1,7	1,9	—		—	
≤50	0,4	1,2	1,8	2,4	3,0	3,4	3,8	—	—		
>50 and ≤125	0,3	0,8	1,1	1,5	1,9	2,1	2,4	—	—	—	
>50 and ≤125	0,3	0,8	1,1	1,5	1,9	2,1	2,4	—		—	
>50 and ≤125	0,6	1,6	2,2	3,0	3,8	4,2	4,8	—	—		
>125 and ≤250	0,6	1,3	1,8	2,5	3,2	3,6	4,0	—	—	—	P
>125 and ≤250	0,6	1,3	1,8	2,5	3,2	3,6	4,0	—	—	—	N/A
>125 and ≤250	1,2	2,6	3,6	5,0	6,4	7,2	8,0	—	—	—	P
>250 and ≤400	1,0	2,0	2,8	4,0	5,0	5,6	6,3	—	—	—	
>250 and ≤400	1,0	2,0	2,8	4,0	5,0	5,6	6,3	—		—	
>250 and ≤400	2,0	4,0	5,6	8,0	10,0	11,2	12,6	—	—		
>400 and ≤500	1,3	2,5	3,6	5,0	6,3	7,1	8,0	—	—	—	
>400 and ≤500	1,3	2,5	3,6	5,0	6,3	7,1	8,0	—		—	
>400 and ≤500	2,6	5,0	7,2	10,0	12,6	14,2	16,0	—	—		
>500 and ≤800	1,8	3,2	4,5	6,3	8,0	9,0	10,0	—	—	—	
>500 and ≤800	1,8	3,2	4,5	6,3	8,0	9,0	10,0	—		—	
>500 and ≤800	3,6	6,4	9,0	12,6	16,0	18,0	20,0	—	—		
>800 and ≤1000	2,4	4,0	5,6	8,0	10,0	11,0	12,5	—	—	—	
>800 and ≤1000	2,4	4,0	5,6	8,0	10,0	11,0	12,5	—		—	
>800 and ≤1000	4,8	8,0	11,2	16,0	20,0	22,0	25,0	—	—		
>1000 and ≤1250	3,2	5,0	7,1	10,0	12,5	14,0	16,0	—	—	—	
>1000 and ≤1250	3,2	5,0	7,1	10,0	12,5	14,0	16,0	—		—	
>1000 and ≤1250	6,4	10,0	14,2	20,0	25,0	28,0	32,0	—	—		
>1250 and ≤1600	4,2	6,3	9,0	12,5	16,0	18,0	20,0		—	—	
>1250 and ≤1600	4,2	6,3	9,0	12,5	16,0	18,0	20,0	—		—	
>1250 and ≤1600	8,4	12,6	18,0	25,0	32,0	36,0	40,0	—	—		
>1600 and ≤2000	5,6	8,0	11,0	16,0	20,0	22,0	25,0		—	—	
>1600 and ≤2000	5,6	8,0	11,0	16,0	20,0	22,0	25,0	—		—	
>1600 and ≤2000	11,2	16,0	22,0	32,0	40,0	44,0	50,0	—	—		
>2000 and ≤2500	7,5	10,0	14,0	20,0	25, 0	28,0	32,0	—	—	—	
>2000 and ≤2500	7,5	10,0	14,0	20,0	25, 0	28,0	32,0	—		—	
>2000 and ≤2500	15,0	20,0	28,0	40,0	50,0	56,0	64,0	—	—		
>2500 and ≤3200	10,0	12,5	18,0	25,0	32,0	36,0	40,0		—	—	
>2500 and ≤3200	10,0	12,5	18,0	25,0	32,0	36,0	40,0	—		—	
>2500 and ≤3200	20,0	25,0	36,0	50,0	64,0	72,0	80,0	—	—		

>3200 and ≤4000	12,5	16,0	22,0	32,0	40,0	45,0	50,0	—	—	—	
>3200 and ≤4000	12,5	16,0	22,0	32,0	40,0	45,0	50,0	—		—	
>3200 and ≤4000	25,0	32,0	44,0	64,0	80,0	90,0	100,0	—	—		
>4000 and ≤5000	16,0	20,0	28,0	40,0	50,0	56,0	63,0	—	—	—	
>4000 and ≤5000	16,0	20,0	28,0	40,0	50,0	56,0	63,0	—		—	
>4000 and ≤5000	32,0	40,0	56,0	80,0	100,0	112,0	126,0	—	—		
>5000 and ≤6300	20,0	25,0	36,0	50,0	63,0	71,0	80,0	—	—	—	
>5000 and ≤6300	20,0	25,0	36,0	50,0	63,0	71,0	80,0	—		—	
>5000 and ≤6300	40,0	50,0	72,0	100,0	126,0	142,0	160,0	—	—		
>6300 and ≤8000	25,0	32,0	45,0	63,0	80,0	90,0	100,0	—	—	—	
>6300 and ≤8000	25,0	32,0	45,0	63,0	80,0	90,0	100,0	—		—	
>6300 and ≤8000	50,0	64,0	90,0	126,0	160,0	180,0	200,0	—	—		
>8000 and ≤10000	32,0	40,0	56,0	80,0	100,0	110,0	125,0	—	—	—	
>8000 and ≤10000	32,0	40,0	56,0	80,0	100,0	110,0	125,0	—		—	
>8000 and ≤10000	64,0	80,0	112,0	160,0	200,0	220,0	250,0	—	—		
>10000 and ≤12500	40,0	50,0	71,0	100,0	125,0	140,0	160,0	—	—	—	
>10000 and ≤12500	40,0	50,0	71,0	100,0	125,0	140,0	160,0	—		—	
>10000 and ≤12500	80,0	100,0	142,0	200,0	250,0	280,0	320,0	—	—		
*), B=Basic, S=Supplementary and R=Reinforced											

29.2	TABLE: Creepage distances, functional insulation							N/A
Working voltage (V)	Creepage distance(mm) Pollution degree							
	1	2			3			
		Material group			Material group			
		I	II	IIIa/IIIb	I	II	IIIa/IIIb	Verdict / Remark
≤50	0,2	0,6	0,8	1,1	1,4	1,6	1,8	
>50 and ≤125	0,3	0,7	1,0	1,4	1,8	2,0	2,2	

>125 and ≤250	0,4	1,0	1,4	2,0	2,5	2,8	3,2	
>250 and ≤400	0,8	1,6	2,2	3,2	4,0	4,5	5,0	
>400 and ≤500	1,0	2,0	2,8	4,0	5,0	5,6	6,3	
>500 and ≤800	1,8	3,2	4,5	6,3	8,0	9,0	10,0	
>800 and ≤1000	2,4	4,0	5,6	8,0	10,0	11,0	12,5	
>1000 and ≤1250	3,2	5,0	7,1	10,0	12,5	14,0	16,0	
>1250 and ≤1600	4,2	6,3	9,0	12,5	16,0	18,0	20,0	

>1600 and ≤2000	5,6	8,0	11,0	16,0	20,0	22,0	25,0	
>2000 and ≤2500	7,5	10,0	14,0	20,0	25,0	28,0	32,0	
>2500 and ≤3200	10,0	12,5	18,0	25,0	32,0	36,0	40,0	
>3200 and ≤4000	12,5	16,0	22,0	32,0	40,0	45,0	50,0	
>4000 and ≤5000	16,0	20,0	28,0	40,0	50,0	56,0	63,0	
>5000 and ≤6300	20,0	25,0	36,0	50,0	63,0	71,0	80,0	
>6300 and ≤8000	25,0	32,0	45,0	63,0	80,0	90,0	100,0	
>8000 and ≤10000	32,0	40,0	56,0	80,0	100,0	110,0	125,0	
>10000 and ≤12500	40,0	50,0	71,0	100,0	125,0	140,0	160,0	

30.1	**TABLE: Ball pressure**			N/A
Part	Test temperature (°C)	Impression diameter (mm)	Allowed impression diameter (mm)	

30.2	**TABLE: resistance to heat, fire and tracking, tracking and glow-wire test**						P
Part under test	Tracking test		Glow wiring test				Result
	175V	250V	550°C	650°C	750 °C	850°C	--
Plastic Enclosure of Switch			√				P

Appendix
Photo documentation

